

CO - 4 ASSESSMENT TOOL -1

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Q17 : Given a set of integers and a target value, determine whether there exists a subset whose sum is exactly equal to the target. Each element can either be included once or excluded from the subset. The objective is to check the existence of at least one valid subset combination. Use Dynamic Programming to solve the problem efficiently.

Input Format:

n = number of elements

arr[] = set elements

sum = target sum

Sample Input:

n = 5

arr = 3 34 4 12 5

sum = 9

Expected Output:

Subset Exists = Yes

Sol :

Source code :

```
def subset_sum(arr, target):  
  
    dp = [False] * (target + 1)  
  
    dp[0] = True  
  
    for x in arr:  
  
        for j in range(target, x - 1, -1):  
  
            dp[j] = dp[j] or dp[j - x]  
  
    return dp[target]  
  
n = 5  
  
arr = [3, 34, 4, 12, 5]  
  
target = 9  
  
if subset_sum(arr, target):
```

```
print("Subset Exists = Yes")
```

else:

```
print("Subset Exists = No")
```

Aim

To determine whether a subset of given integers has a sum exactly equal to the target value.

Input

- n = number of elements
- $arr[]$ = set of integers
- $target$ = required sum

Output

Yes if a valid subset exists, otherwise No.

Algorithm

1. Create a DP array initialized to False.
2. Set $dp[0] = \text{True}$.
3. For each element, update possible sums in reverse order.
4. If $dp[target]$ becomes True, a valid subset exists.

Example

```
arr = [3, 34, 4, 12, 5]  
target = 9
```

Valid subset:

$$4 + 5 = 9$$

Complexity

Time: $O(n \times target)$

Space: $O(target)$

Result

The Dynamic Programming approach successfully determines that a subset with sum **9** exists.

Output Screenshot :

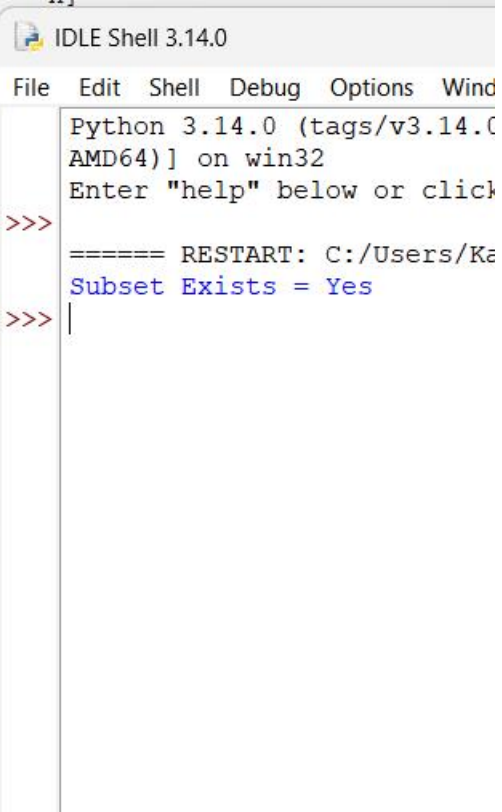
```
def subset_sum(arr, target):
    dp = [False] * (target + 1)
    dp[0] = True

    for x in arr:
        for j in range(target, x - 1, -1):
            dp[j] = dp[j] or dp[j - x]

    return dp[target]

n = 5
arr = [3, 34, 4, 12, 5]
target = 9

if subset_sum(arr, target):
    print("Subset Exists = Yes")
else:
    print("Subset Exists = No")
```



The screenshot shows the IDLE Shell 3.14.0 interface. The left pane contains the Python code for the subset_sum function and its execution. The right pane shows the output of the program, which is "Subset Exists = Yes". The output is displayed in a blue font, matching the color of the function name in the code.